

Market and User Research Report: Twi Audio Transcriber

1. Executive Summary & Project Origin

The Twi Audio Transcriber is a translation-first accessibility platform designed to bridge communication barriers between Asante Twi speakers and individuals who do not speak or fully understand the language.

The project was inspired by the lived experience of a non-Twi-speaking professional living and working in Ghana. In both personal and professional settings, such individuals frequently encounter audio recordings, verbal instructions, meetings, and discussions conducted entirely in Twi. While these recordings often contain important information, understanding them can be difficult and may require repeatedly replaying audio or seeking assistance from fluent speakers.

The Twi Audio Transcriber seeks to address this challenge by transforming spoken Twi recordings into readable and understandable English text. Rather than focusing solely on speech recognition technology, the project focuses on communication accessibility—helping users access, understand, and act upon information that would otherwise be inaccessible due to language barriers.

The platform is designed to process a wide variety of audio recordings, including:

- Interviews
- Meetings
- Lectures
- Training sessions
- Public speeches
- Community discussions
- Recorded instructions
- Podcasts and broadcasts

The ultimate goal is to improve communication, inclusivity, and information accessibility within Ghana's multilingual environment.

2. Market Landscape & Competitive Analysis

Research into Ghana's language technology ecosystem reveals several organizations working in speech recognition and translation. However, most existing solutions primarily focus on language technology infrastructure, enterprise services, or institutional archiving rather than solving everyday communication challenges faced by non-native speakers.

Intron Health (Sahara-v2)

Intron Health provides speech recognition infrastructure designed for African languages and accents, particularly within healthcare and enterprise environments. Their offering is geared toward organizations and developers seeking voice AI integrations rather than individual users requiring day-to-day communication assistance.

Sesa AI

Sesa AI focuses on professional transcription, translation, and digital archiving. The platform is particularly suited for formal environments such as legal proceedings, corporate documentation, media archiving, and institutional record-keeping. Its primary value proposition centers on structured transcription workflows and compliance.

3. Competitive Matrix

Feature / Criteria	Intron Health (Sahara-v2)	Sesa AI	Twí Audio Transcriber
Primary Audience	Enterprises, healthcare institutions, developers	Legal, media, and corporate organizations	Non-Twi-speaking professionals, students, residents, and expatriates
Core Value Proposition	African language speech recognition infrastructure	Formal transcription and archiving services	Communication accessibility between Twi speakers and non-speakers
Output Focus	Speech-to-text transcription	Formal transcription and translation	English-first understanding and information accessibility
User Experience	API and enterprise-focused	Institutional workflows and compliance	Lightweight, intuitive, and user-friendly
Speech Type	General speech recognition	Structured and formal speech	Everyday conversations, instructions, and informal communication
Deployment Model	Enterprise SaaS and APIs	Subscription-based business platform	Direct consumer-focused application

4. Key Differentiators

The Twí Audio Transcriber occupies a unique position within the market through four key differentiators.

1. Translation-First Workflow

Many existing solutions follow a two-step process where speech is first converted into Twi text before being translated into English.

For users who cannot read or understand Twi, this intermediate step provides little value.

The Twi Audio Transcriber is designed with English understanding as the primary outcome. While the system may internally generate Twi transcripts for validation purposes, the main user-facing output is clear and understandable English text.

This approach minimizes friction and allows users to access information immediately.

2. General Audio Recording Support

Unlike solutions optimized for specific use cases such as legal proceedings or archived records, the Twi Audio Transcriber is designed to work with audio recordings in general.

Supported scenarios include:

- Workplace meetings
- Interviews
- Lectures
- Community discussions
- Training materials
- Spoken instructions

This versatility makes the platform useful across both personal and professional contexts.

3. Optimization for Informal Speech and Code-Switching

Daily communication in Ghana frequently involves a mixture of Twi and English, commonly referred to as code-switching.

Many formal transcription systems are optimized for structured speech and may struggle with conversational language, local expressions, and rapid informal communication.

The Twi Audio Transcriber is positioned to handle the realities of everyday spoken interactions, making it more relevant to its target audience.

4. Communication Accessibility Rather Than Language Processing

Most existing solutions focus on language technology capabilities such as speech recognition, translation engines, or transcription services.

The Twi Audio Transcriber focuses on a different problem:

How can people who do not understand Twi access and use information contained in spoken recordings?

The project's primary objective is not simply converting speech into text. Instead, it seeks to improve communication, accessibility, and inclusion by making spoken information understandable across language barriers.

5. Long-Term Vision and Future Expansion

While the initial focus is on Twi-to-English transcription and translation, the broader vision extends beyond a single language.

The platform could evolve into a Ghanaian Language Accessibility Platform supporting:

- Twi
- Ga
- Ewe
- Dagbani
- Hausa
- Other local languages

This expansion would further strengthen communication and accessibility in Ghana's multilingual society, helping users overcome language barriers regardless of the language involved.

6. Conclusion

The Twi Audio Transcriber is not merely a speech-to-text application. It is a communication accessibility platform designed to help non-Twi speakers understand and utilize information contained within spoken Twi recordings.

By focusing on user understanding rather than language processing alone, the project occupies a distinct position within the Ghanaian language technology ecosystem and addresses a practical challenge experienced by thousands of residents, students, professionals, and expatriates across the country.

Its long-term vision is to foster a more inclusive and connected society where language differences no longer prevent people from accessing important information.